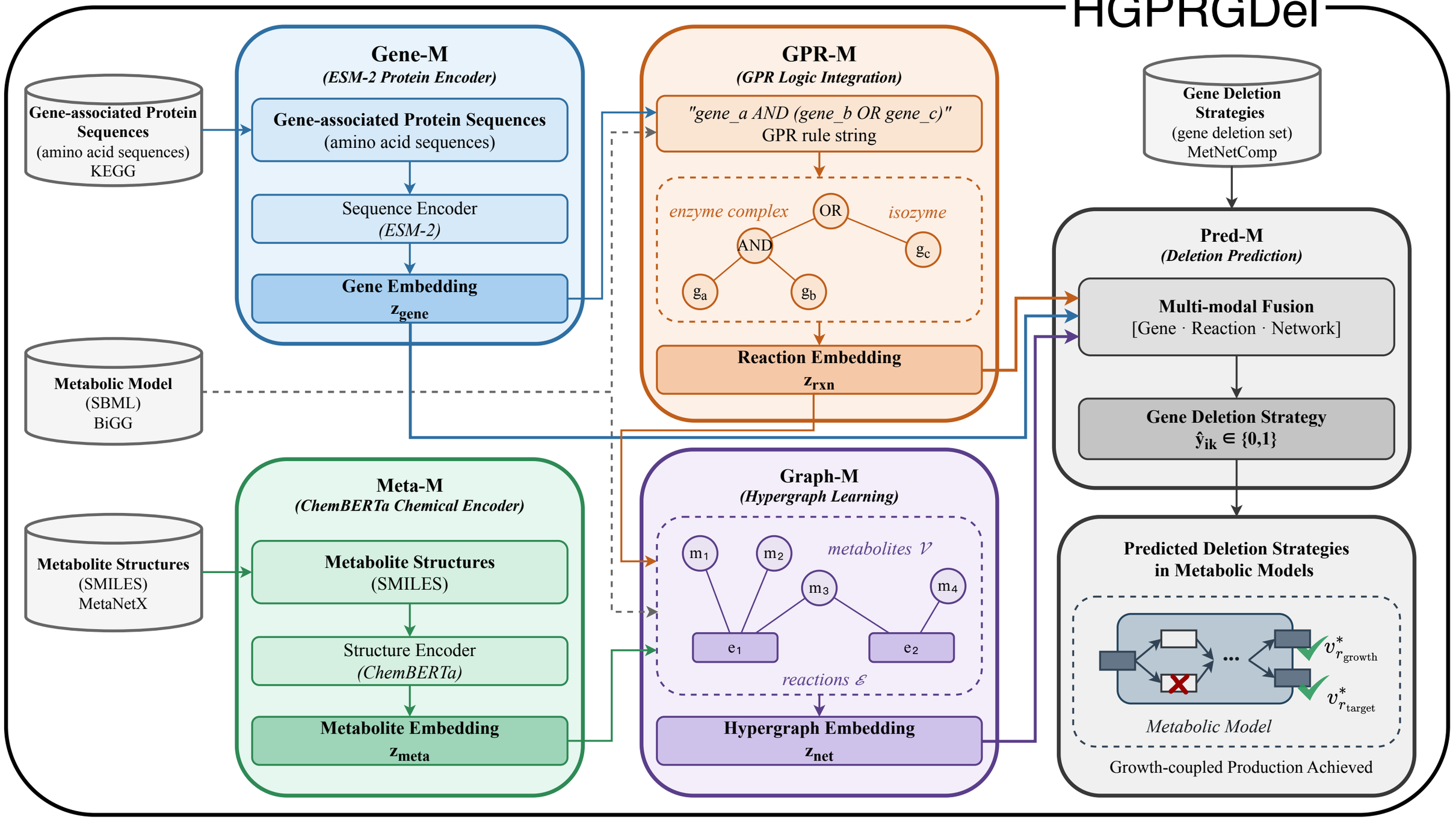

HGPRGDel
Gene-M
(ESM-2 Protein Encoder)
Gene-associated Protein Sequences
(amino acid sequences)
KEGG
Gene-associated Protein Sequences
(amino acid sequences)
Sequence Encoder
(ESM-2)
Gene Embedding
z_{gene}
GPR-M
(GPR Logic Integration)
"gene_a AND (gene_b OR gene_c)"
GPR rule string
enzyme complex
OR
isozyme
AND
g_c
g_a
g_b
Reaction Embedding
z_{rxn}
Gene Deletion Strategies
(gene deletion set)
MetNetComp
Pred-M
(Deletion Prediction)
Multi-modal Fusion
[Gene · Reaction · Network]
Gene Deletion Strategy
$\hat{y}_{ik} \in \{0,1\}$
Metabolic Model
(SBML)
BiGG
Meta-M
(ChemBERTa Chemical Encoder)
Metabolite Structures
(SMILES)
MetaNetX
Metabolite Structures
(SMILES)
Structure Encoder
(ChemBERTa)
Metabolite Embedding
z_{meta}
Graph-M
(Hypergraph Learning)
m_1
m_2
metabolites $\mathcal{V}$
m_3
m_4
e_1
e_2
reactions $\mathcal{E}$
Hypergraph Embedding
z_{net}
Predicted Deletion Strategies
in Metabolic Models
$v^*_{r_{growth}}$
$v^*_{r_{target}}$
Metabolic Model
Growth-coupled Production Achieved